CLASSWORK 8

Analysis in Linear Regression

CPE 486/586

Instructor: Rahul Bhadani **Due: September 30, 2026, In-Class**

20 points

1. In a linear model $g(x) = \mathbf{w}^\top \mathbf{x} + w_0$, what role does $\mathbf{x}$ play? **(2 Points)**

Answer

2. What assumptions we make about model in simple linear regression? **(2 Points)**

Answer

3. What do we understand by least square fit in linear regression? **(2 Points)**

Answer

4. You are given a dataset about a traffic that contains information about traffic density (in vehicles/miles) and traffic flow (vehicles/hour). You are required to create a machine learning that let you predict traffic flow for a given traffic density. How will you proceed with creating such model and then make a prediction about traffic flow from a given traffic density value? **(8 Points)**

Answer

5. What does coefficient of determination R^2 tells about a linear regression model? **(2 Points)**

Answer

6. You are given two independent random variable X and Y that have exhibited same populaton variance σ^2 and it is known to us. Assuming their respective sample variances are s_X^2 and s_Y^2 when they each have m and n samples.

   (a) What is the difference between population variance σ^2 and sample variance s_X^2? **(2 Points)**

   Answer

   (b) What probability distribution the random quantity $\frac{s_Y^2}{s_X^2}$ exhibit? **(2 Points)**

   Answer